

SHORT ANSWER TEST

- 1) $E \rightarrow E + E \mid id$ $\Rightarrow$ ambiguous or not.
- 2) Construct NFA for $a(a|b)^*a$.
- 3) Construct parse tree for $id + id * id$.
- 4) Eliminate left recursion for $A \rightarrow A\alpha \mid \beta$.

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$$A \rightarrow A\alpha \mid \beta$$

$$A \rightarrow A\alpha$$

$$A \rightarrow \beta$$

$$A' \rightarrow \beta A'$$

$$A' \rightarrow \alpha A' \mid \epsilon$$

$$(3) \quad id + id * id$$

$$E \rightarrow E$$



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- 5) $S \rightarrow iEtS \mid iEtSes \mid a$ - Left factoring

$$S \rightarrow iEtS$$

$$S \rightarrow iEtSes$$

$$S \rightarrow a$$

Formula

$$iEtS \rightarrow a$$

(ii) Left factoring:

$$S \rightarrow iEtSes'$$

$$S \rightarrow iEs$$

6) $S \rightarrow aBDh \Rightarrow$ calculate first & follow

2. $B \rightarrow cC$

3. $C \rightarrow bC/\epsilon$

4. $D \rightarrow EF$

5. $E \rightarrow g/\epsilon$

6. $F \rightarrow f/\epsilon$

(i) No left recursion needed

(ii) FIRST():-

1. $FIRST(S) \Rightarrow \{a\}$

2. $FIRST(B) \Rightarrow \{c\}$

3. $FIRST(C) \Rightarrow \{b, \epsilon\}$

4. $FIRST(D) \Rightarrow \{g, f\}$

5. $FIRST(E) \Rightarrow \{g, \epsilon\}$

6. $FIRST(F) \Rightarrow \{f, \epsilon\}$

(iii) FOLLOW():-

1. $FOLLOW(S) = \{\$, \}$

2. $FOLLOW(B) = \{\$, \}$

3. $FOLLOW(C) = \{\$, \}$

4. $FOLLOW(D) = \{\$, \}$

5. $FOLLOW(E) = \{g, f, \$\}$

6. $FOLLOW(F) = \{g, f, \$\}$

$S \rightarrow aBDh$
 $\alpha B \beta$

$B \rightarrow cC$
 $\alpha B \beta$

$C \rightarrow bC/\epsilon$
 $\alpha B \beta$

$D \rightarrow EF$
 $\alpha B \beta$

$E \rightarrow g/\epsilon$
 $\alpha \beta$

$$7> S \rightarrow Sa | Sb | cd$$

$$S \rightarrow Sa$$

$$S \rightarrow Sb$$

$$S \rightarrow cd$$

No, immediate left recursion
does not occurs.

$$8> E \rightarrow TE'$$

$$E' \rightarrow +TE' | \epsilon, \text{ find } \text{FIRST}(E')$$

$$\text{FIRST}(E') \Rightarrow \{+, \epsilon\}$$

9> In the grammar $S \rightarrow (S)S | \epsilon$, the string $(())$ is
invalid.

10> Shift - Reduce parsing:

There are 2 actions that are mainly performed
in Shift-Reduce Parsing and they are

(i) ~~closure~~ ^{closure} & (ii) Goto

$$11> \text{Grammar: } S' \rightarrow s$$

$$S \rightarrow cc$$

$$c \rightarrow cc | a$$

$$[\text{Item: } S' \rightarrow \cdot s]$$

108 Shift - Reduce parsing:

In Shift - Reduce parsing

117 Grammar: $S' \rightarrow S$
 $S \rightarrow CC$
 $C \rightarrow cC \mid d$

closure I_0 : $S' \rightarrow \cdot S$
 $S \rightarrow \cdot CC$
 $C \rightarrow \cdot cC \mid \cdot d$

124 (i) $S \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow b$

$S \rightarrow \begin{matrix} AB \\ aB \end{matrix}$

FIRST:

$\leftarrow \text{FIRST}(S) \rightarrow$

(ii) For the grammar $aabb$ is accepted.

$S \rightarrow asb \mid ab$, check whether

Sol:

(i) $S \rightarrow AB$
 $A \rightarrow a$
 $B \rightarrow b$

$S \rightarrow asb \mid ab$

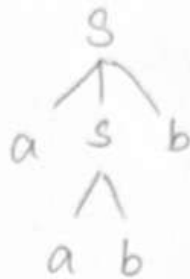
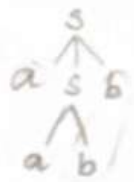
$\begin{matrix} S \\ / \end{matrix}$

1. $\text{FIRST}(S) = \{a\}$
2. $\text{FIRST}(A) = \{a\}$
3. $\text{FIRST}(B) = \{b\}$

(ii) $S \rightarrow aSb \mid ab$, (aabb)

$S \rightarrow aSb$

$S \rightarrow ab$



the string "aabb" is ^{not} accepted by using
 parse tree.

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$S \rightarrow A$

$A \rightarrow aB \mid Ad$

$B \rightarrow b$

$C \rightarrow g$

} $\Rightarrow$ FIRST & FOLLOW

$A \rightarrow aBA'$
 $A' \rightarrow \epsilon \mid dA'$

Sol:-

FIRSTLY:

(a) Left recursion:

1. $S \rightarrow A$

2. $A \rightarrow aB \mid Ad$

$A \rightarrow aB$

$A \rightarrow Ad$

$A \rightarrow aBA'$

$A' \rightarrow dA' \mid \epsilon$

3. $B \rightarrow b$

4. $C \rightarrow g$

Grammar after
 left recursion:

1. $S \rightarrow A$

2. $A \rightarrow aBA'$

3. $A' \rightarrow dA' \mid \epsilon$

4. $B \rightarrow b$

5. $C \rightarrow g$